

Snakes

PHYLUM	Chordata
CLASS	Reptilia
ORDER	Squamata
SUBORDER	Serpentes
FAMILIES	27
SPECIES	About 4,500

CLASSIFICATION NOTE

Snakes are closely related to lizards and amphisbaenians. There is much debate over the relationships between different types of snakes, and there is no universally accepted system of classification. Most controversy concerns the number of families. While most authorities recognize 19, others consider some of these to be subfamilies and arrive at a smaller number, combining boas and pythons in a single family, for example.

Snakes are formidable and highly evolved predators. Although they have no limbs, no eyelids, and no external ears, these versatile animals move with ease and find their prey using sophisticated senses. All snakes eat other animals, ranging from ants to antelopes, some of which they subdue by constriction or by delivering a venomous bite from specialized fangs. Although snakes cannot chew their food, the bones in their skull are lightly constructed and loosely connected, so that the jaws can be opened wide and the prey swallowed whole. Snakes have established themselves on all the world's major landmasses (except Antarctica) as well as on many oceanic islands. Only about one in 10 are venomous and of these only a small proportion represent a threat to humans.

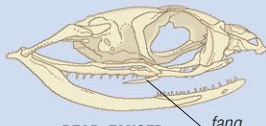
Anatomy

The shape of a snake's body usually reflects where it lives: climbing species tend to be long and thin; burrowers are often short and stout with short tails and blunt snouts; and sea snakes have flattened, paddle-shaped tails. Unlike other reptiles, snakes have a single row of scales on their underside, the ventral scales, which are usually wide, and smaller scales on the upper surfaces. Some species have large, regular plates on the head, while others have small, fragmented scales. Each eye is covered with a transparent scale, the brille or spectacle, which is replaced when the snake sheds its skin. All scales can be smooth, ridged (keeled), or granular. The internal organs are modified to fit the elongated body. Paired organs are staggered within the body cavity, and only one lung may be functional, with the other reduced in size. Sea snakes have an enlarged lung, part of which forms a buoyancy chamber.



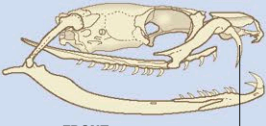
PRIMITIVE SNAKE

short, heavy jaw



REAR-FANGED SNAKE

fang at rear, below eye



FRONT-FANGED SNAKE

fixed hollow fang

SKULLS AND TEETH

Primitive snakes have a heavy skull with few teeth. Most other snakes have a lighter skull with loosely connected jawbones that can move apart. Teeth are fixed to the upper or lower jaw, or to the roof of the mouth. Venom-injecting fangs may be at the front or rear of the mouth. In some front-fanged snakes, they are hinged to the upper jaw.



SKELETON

A snake has up to 400 vertebrae that articulate on each other to give a highly flexible skeleton. Each vertebra has 2 winglike processes to keep the spine from twisting. Ribs are attached to the vertebrae in the body but are absent in the tail. Some species have hind limb girdles, and primitive snakes have vestigial hind limbs in the form of small spurs, but no snakes have front-limb girdles.

SENSES

Snakes have poor eyesight and hearing, and rely instead on other senses. A well-developed sense of smell is supplemented by the Jacobson's organ (see p.373), and active snakes flick their tongue constantly to sample their surroundings. Pit vipers and some boas and pythons can detect small changes in air temperature using organs on their face known as heat pits.



HEAT PITS



FORKED TONGUE

SWALLOWING A LIZARD



1

STEALTHY APPROACH

A vine snake slides slowly and carefully along a branch toward an unsuspecting gecko.



2

THE ATTACK

In one swift movement, the snake arches over the gecko and, using its rear fangs, injects its venom.



3

HOLDING ON

Although the gecko falls from its perch, the snake's tail is wrapped around a branch and it holds on.



4

SHIFTING GRIP

Once the gecko is dead, the snake begins to maneuver it into position for swallowing.